In [ ]:

```
import pandas as pd
dataset=pd.read_csv("Startup_dataset.csv")
dataset
```

In [4]:

```
dataset=dataset.drop(columns=["Unnamed: 0"])
dataset
```

Out[4]:

	RD Spend	Administration	Market Spend	State	Profit
0	165349.30	136897.90	471784.20	New York	192261.93
1	162597.80	151377.69	443898.63	California	191792.16
2	153441.61	101145.65	407934.64	Florida	191050.49
3	144372.51	118671.95	383199.72	New York	182902.09
4	142107.44	91391.87	366168.52	Florida	166188.04
5	131877.00	99814.81	362861.46	New York	156991.22
6	134615.56	147198.97	127716.92	California	156122.61
7	130298.23	145530.16	323876.78	Florida	155752.70
8	120542.62	148719.05	311613.39	New York	152211.87
9	123334.98	108679.27	304981.72	California	149760.06
10	101913.18	110594.21	229161.05	Florida	146122.05
11	100672.06	91790.71	249744.65	California	144259.50
12	93863.85	127320.48	249839.54	Florida	141585.62
13	91992.49	135495.17	252665.03	California	134307.45
14	119943.34	156547.52	256513.02	Florida	132602.75
15	114523.71	122616.94	261776.33	New York	129917.14
16	78013.21	121597.65	264346.16	California	126993.03
17	94657.26	145077.68	282574.41	New York	125370.47
18	91749.26	114175.89	294919.67	Florida	124267.00
19	86419.80	153514.21	0.10	New York	122776.96
20	76253.96	113867.40	298664.57	California	118474.13
21	78389.57	153773.53	299737.39	New York	111313.12
22	73994.66	122782.85	303319.36	Florida	110352.35
23	67532.63	105751.13	304768.83	Florida	108734.09
24	77044.11	99281.44	140574.91	New York	108552.14
25	64664.81	139553.26	137962.72	California	107404.44
26	75328.97	144136.08	134050.17	Florida	105733.64
27	72107.70	127864.65	353183.91	New York	105008.41

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	RD Spend	Administration	Market Spend	State	Profit
28	66051.62	182645.66	118148.30	Florida	103282.48
29	65605.58	153032.16	107138.48	New York	101004.74
30	61994.58	115641.38	91131.34	Florida	99937.69
31	61136.48	152702.02	88218.33	New York	97483.66
32	63408.96	129219.71	46085.35	California	97427.94
33	55494.05	103057.59	214634.91	Florida	96779.02
34	46426.17	157694.02	210797.77	California	96712.90
35	46014.12	85047.54	205517.74	New York	96479.61
36	28663.86	127056.31	201126.92	Florida	90708.29
37	44070.05	51283.24	197029.52	California	89949.24
38	20229.69	65948.03	185265.20	New York	81229.16
39	38558.61	82982.19	174999.40	California	81005.86
40	28754.43	118546.15	172795.77	California	78240.01
41	27893.02	84710.87	164470.81	Florida	77798.93
42	23641.03	96189.73	148001.21	California	71498.59
43	15505.83	127382.40	35534.27	New York	69759.08
44	22177.84	154806.24	28334.82	California	65200.43
45	1000.33	124153.14	1904.03	New York	64926.18
46	1315.56	115816.31	297114.56	Florida	49490.85
47	0.10	135427.02	0.10	California	42559.83
48	542.15	51743.25	0.10	New York	35673.51
49	0.10	116983.90	45173.16	California	14681.50

In [20]:

```
x=dataset[["RD Spend","Administration","Market Spend"]]
y=dataset[["Profit"]]
x
y
```

Out[20]:

	Profit
0	192261.93
1	191792.16
2	191050.49
3	182902.09
4	166188.04
5	156991.22
6	156122.61

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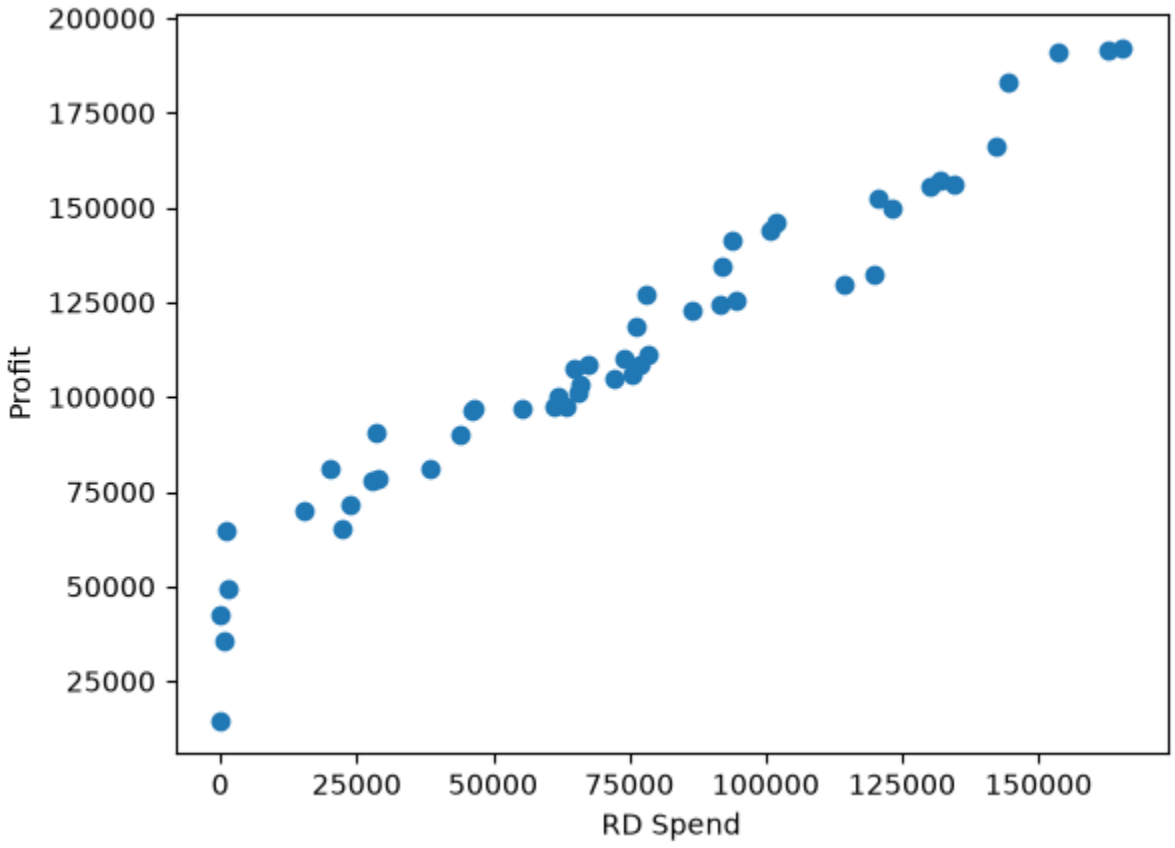
	Profit
7	155752.70
8	152211.87
9	149760.06
10	146122.05
11	144259.50
12	141585.62
13	134307.45
14	132602.75
15	129917.14
16	126993.03
17	125370.47
18	124267.00
19	122776.96
20	118474.13
21	111313.12
22	110352.35
23	108734.09
24	108552.14
25	107404.44
26	105733.64
27	105008.41
28	103282.48
29	101004.74
30	99937.69
31	97483.66
32	97427.94
33	96779.02
34	96712.90
35	96479.61
36	90708.29
37	89949.24
38	81229.16
39	81005.86
40	78240.01
41	77798.93

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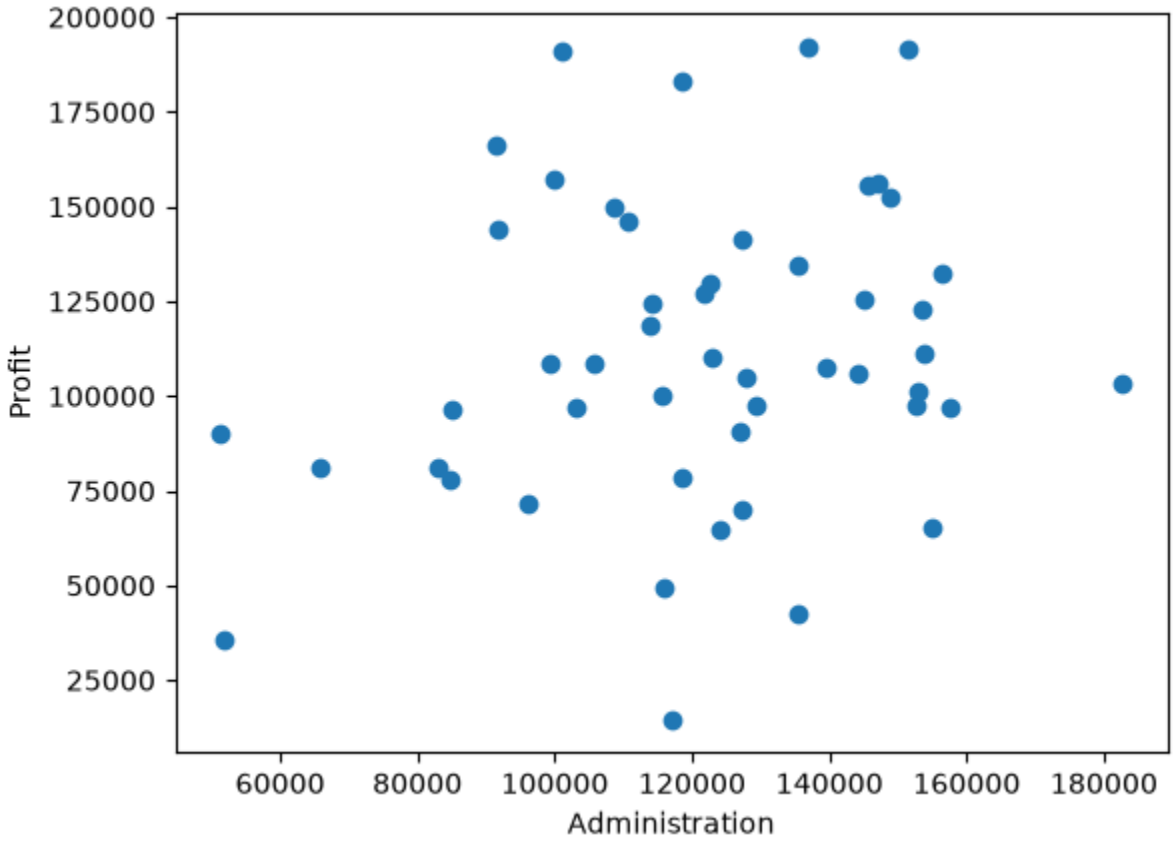
	Profit
42	71498.59
43	69759.08
44	65200.43
45	64926.18
46	49490.85
47	42559.83
48	35673.51
49	14681.50

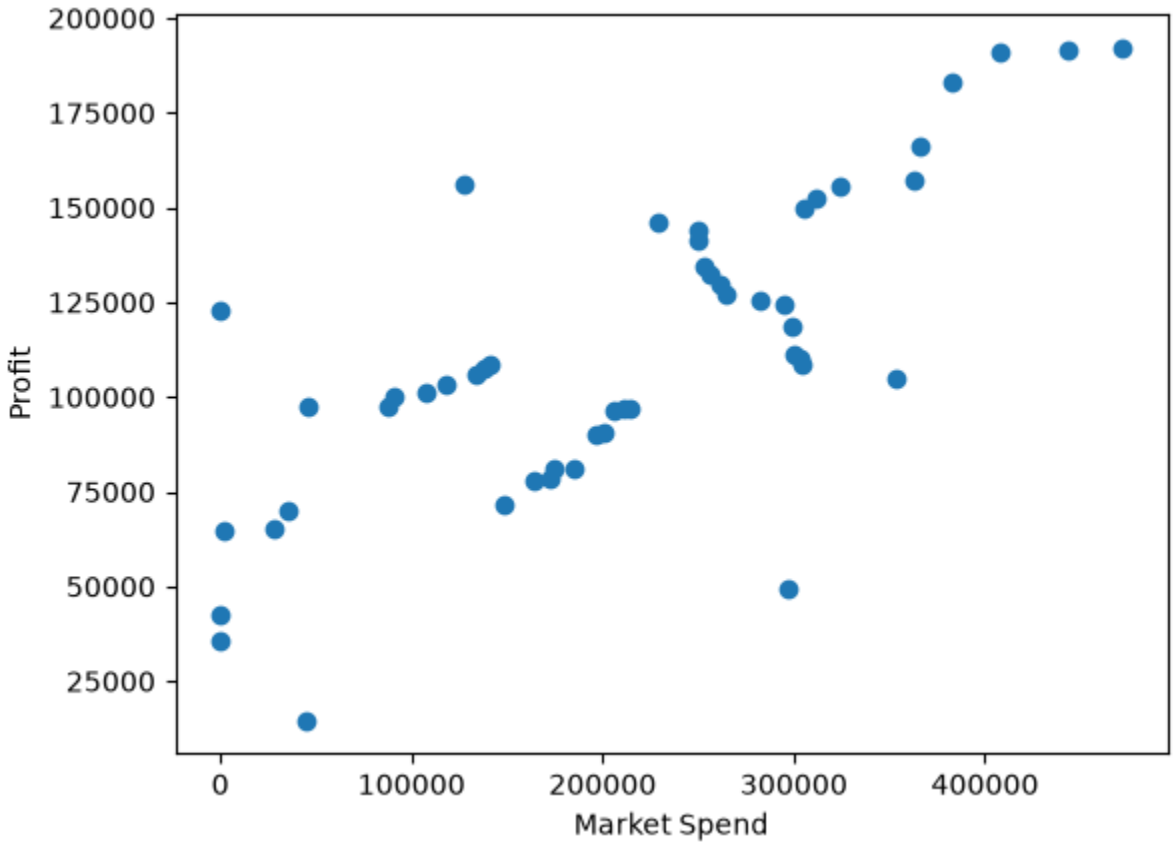
In [12]:

```
import matplotlib.pyplot as plt
plt.scatter(dataset["RD Spend"], y)
plt.xlabel("RD Spend")
plt.ylabel("Profit")
plt.show()
plt.scatter(dataset["Administration"], y)
plt.xlabel("Administration")
plt.ylabel("Profit")
plt.show()
plt.scatter(dataset["Market Spend"], y)
plt.xlabel("Market Spend")
plt.ylabel("Profit")
plt.show()
```



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In [30]:

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.2,random_state=42)
y_test
x_test
```

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Out[30]:

	RD Spend	Administration	Market Spend
13	91992.49	135495.17	252665.03
39	38558.61	82982.19	174999.40
30	61994.58	115641.38	91131.34
45	1000.33	124153.14	1904.03
17	94657.26	145077.68	282574.41
48	542.15	51743.25	0.10
26	75328.97	144136.08	134050.17
25	64664.81	139553.26	137962.72
32	63408.96	129219.71	46085.35
19	86419.80	153514.21	0.10

In [28]:

```
from sklearn.linear_model import LinearRegression
reg=LinearRegression()
reg.fit(x_train, y_train)
```

Out[28]:

▾ LinearRegression ⓘ ⓘ

▸ Parameters

▸ Fitted attributes

In [33]:

```
y_pred =reg.predict(x_test)
print("Predicted Profit:")
print(y_pred)
print("actual profit:")
print(y_test.values)
```

```
Predicted Profit:
[[126703.12716461]
 [ 84894.85081556]
 [ 98893.51815974]
 [ 46501.80815036]
 [129128.49734381]
 [ 50992.79486261]
 [109016.6536578 ]
 [100878.5641454 ]
 [ 97700.69638629]
 [113106.25292226]]
actual profit:
[[134307.45]
 [ 81005.86]
 [ 99937.69]
 [ 64926.18]
```

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```
 [125370.47]
 [ 35673.51]
 [105733.64]
 [107404.44]
 [ 97427.94]
 [122776.96]]
```

In [34]:

```
from sklearn.metrics import r2_score
result=r2_score(y_pred,y_test)
result
```

Out[34]:

```
0.8865921975655436
```

In [37]:

```
pro=reg.predict([[48,60,75]])
pro
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning:
X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Out[37]:

```
array([[54108.74780756]])
```

In [38]:

```
m=reg.coef_
m
```

Out[38]:

```
array([[ 0.80377928, -0.06792917,  0.03124155]])
```

In [39]:

```
c=reg.intercept_
c
```

Out[39]:

```
array([54071.89903683])
```

In [40]:

```
import pickle
filename="SML_model.sav"
pickle.dump(reg, open(filename,"wb"))
```

In [ ]:

```

```

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